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Optimal Dynamic Resource Allocation for Multi-Point Communication in WBAN

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ABSTRACT Considering the characteristics of lower power resources in multi-point wireless body area networks (WBANs), the wireless energy and information transmission in full duplex (FD) mode is considered. We propose a transmission protocol based on time division multiple access, which includes two phases. In the inactive phase, the signal access point (AP) broadcasts Radio Frequency (RF) energy to the sensors. When the energy collected by all sensors reaches their respective energy thresholds, the active phase will start. In the active phase, the sensors send back the collected physiological information parameters to the AP, while sensors still collecting the energy transmitted from the AP to achieve the purpose of energy accumulating. Our goal is to obtain the optimal time allocation ratio through the joint objective linear programming method to obtain the maximum average sum-throughput. The simulation results compared with the AP with FD and the sensors with half duplex (HD) show that our proposed optimization method can improve system throughput obviously.

INDEX TERMS Wireless body area network (WBAN), energy harvest, energy and information transfer, full duplex (FD), half duplex (HD), linear programming method.

I. INTRODUCTION

With the development of wireless communication technology and intelligent electronic sensors, various electronic health applications have come into our sight. Wireless body area networks (WBANs) are dynamic heterogeneous networks including sensors that have different sizes, uses, energy demands, etc [1]. WBAN can monitor physical conditions. Early diagnosis is essential for the diagnosis and treatment of the disease [2]. The sensors in WBAN are usually battery-powered, however, battery replacement is difficult for implantable devices. Therefore, in order to improve the life time cycle, it is particularly important to solve the power problem of the sensor [3].

In recent years, the emergence of radio frequency (RF) energy collection technology is an effective way to solve

the energy limitation of wireless networks [4]. RF signals are used to transmit wireless energy and information [5]. In [6], the authors propose wireless power transfer (WPT) to collect energy through electromagnetic radiation. However, due to concerns about human health and the stability of collected energy in WBAN, we mainly consider RF energy collection technology that can provide stable and continuous energy. Simultaneous wireless information and power transfer (SWIPT) [7] is proposed for delivering RF energy. In [8], the authors propose dynamic power splitting (DPS), study three special cases of DPS, and propose separated and integrated receiver architectures to realize the decoding of the information signal carried from the energy signal. Authors in [9] mainly summarize three energy harvester models, and describe the difference between the energy harvester applied to a single user and multiple users. In WBAN, the sensor usually collects energy and then transmits information. Most papers discuss the communication between single points.

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This paper studies the information transmission of the point-to-point communication system in WBAN when the system is in normal and abnormal states respectively. Different protocols are used to obtain and solve the optimal problem of maximum throughput [10]. In [11], a single-point communication network with relay is studied, and the throughput of information transmission under time splitting (TS) and power splitting (PS) protocols is verified respectively. On this basis, an adaptive maximum ratio combining (MRC) protocol is proposed. In [12], the authors study the control strategy of optimizing time and power simultaneously, and optimize the duty cycle and transmission power to achieve the optimal state.

In WBAN, due to its characteristics of high dynamics and complexity, the channel varies with the position of the sensor, so the communication between single points can not meet the system requirements. Therefore, the transmission of energy and information between multiple points is gradually studied. In [13], they study two cases of system normal and system abnormal in multi-point wireless body area network (MP-WBAN), and propose a hybrid TS and PS protocol, using convex optimization technology and iterative algorithm to get the maximum throughput solution. The model in [14] is a multi-point and an access point (AP) with multi-antenna system. The authors consider the path loss and shadow value, and use the maximum ratio transmission (MRT) and zero-forcing (ZF) techniques to make the system energy-saving and reliable. In [15], under the premise of ensuring the quality of service of each user, the SWIPT beamforming design problem of multiple cells and multiple users is studied, and the design methods of centralized and decentralized algorithms are proposed respectively.

In addition, in order to improve SWIPT transmission efficiency, relay systems and full duplex (FD) systems are gradually proposed. As for relay systems, the authors in [16] propose two relaying protocols, namely, time switching-based relaying (TSR) protocol and power splitting-based relaying (PSR) protocol to enable energy harvesting and information processing at the relay. In [17], the authors study a bidirectional information transmission model with a relay, and compares the system performance in continuous and in discrete time. The authors in [18] study the multi-point with a relay system in WBAN. By allocating the power at the relay, the authors divide the optimization problem into destination node power limit (DPL) and relay power limit (RPL), and finally solve it by Lagrange multiplier method. In massive multiple-input-multiple-output (MIMO), the transmission distance is short, and the authors theoretically derive a gradual expression of the rate when there are many transmitting antennas [19]. The authors in [20] investigate the resource allocation and relay selection in a two-hop relay-assisted multi-user orthogonal frequency division multiple access (OFDMA) network, where the end nodes support SWIPT employing a PS technique.

As for FD systems, the authors study the joint design of beamforming matrix in multi-user FD system. By using

semidefinite relaxation (SDR) method, the solution of self-interfering channel is obtained [21]. In [22], the authors propose a FD wireless-powered communication network (FD-WPCN), and design a protocol based on time allocation. They jointly optimize the time and power allocation of uplink (UL) and downlink (DL) communications to maximize the user weighted sum rate. In [23] a two-stage protocol is proposed in a full-duplex network with a relay to improve the performance of information transmission, and also enable self-energy recovery. In [24], the authors design a transceiver structure, so that each user can receive energy and information at the same time under FD, which improves the system throughput. In [25], artificial noise (AN) is used to realize the anti-eavesdropping SWIPT in the FD system, and the weight is improved by jointly optimizing the information covariance matrix and the receiving vector.

In this paper, we consider a FD multipoint system for simultaneous transmission of energy and information in WBAN. Both the AP and sensors working in half duplex (HD) mode in WBAN have the problems of low resource utilization and high delay. WBANs need high delay performance of the system, in order to effectively utilize system resources and improve the performance of data transmission. At the same time, considering that there are many kinds of physiological parameters in the human body, and a single sensor can not obtain many parameters, so we study the communication between multiple points, which is more in line with the actual needs. We assume that both AP node and sensors work in FD mode. It is assumed that the AP node has sufficient energy, the sensors' energy all come from the AP node, and the sensors have no initial energy. Similar transmission systems have been described in several papers. In [22], energy can not be collected at the same time in the information transmission phase of the sensor, and energy and information are not transmitted at the same time. In [24], although the simultaneous transmission of energy and information is realized, the sensors have initial energy, which is not considered when the sensor has no initial energy supply. Different from previous work, in this paper, we consider that the sensors have no initial energy and all sensors and the AP work in FD mode. Moreover, the simultaneous transmission of energy and information is realized in the transmission process. The main contributions of this paper are summarized as follows:

- According to the characteristics of the multipoint WBAN system model, we consider that all sensors and the AP work in FD mode, and the sensors acquire energy from the RF signals emitted by the AP. After the collected energy of all sensors reaches the threshold value, the physiological parameter information is transmitted to the AP while the sensors also harvest energy from the AP.
- According to the channel characteristics of multipoint WBAN, a transmission protocol based on time allocation is designed. The whole block stage is divided into active and inactive stages. In the active stage, we realize

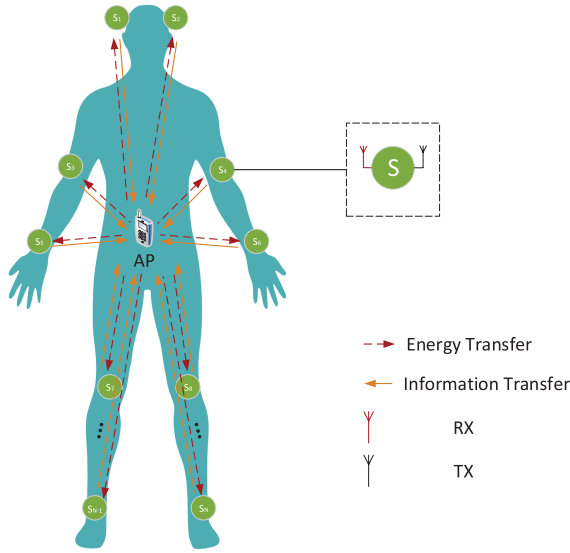


FIGURE 1. Multipoint FD SWIPT in WBAN.

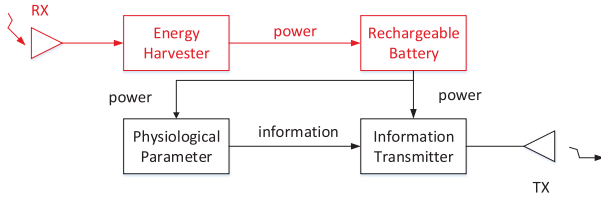


FIGURE 2. The structure of FD sensors.

the simultaneous transmission of energy and information and the energy accumulating.

- Aiming at the average sum-throughput maximization problem, the corresponding optimization algorithm is proposed to obtain the optimal time allocation ratio. At the same time, the performance of the system is verified by comparing AP in FD mode and sensors in HD mode.

The rest of this paper is organized as follows. In section II, the system model and the FD transmission protocol are introduced. In section III, the optimization strategy to get the optimal average throughput is studied. A lot of simulation results are displayed in section IV to show the performance of the proposed algorithms. Section V is the conclusion of the paper.

II. SYSTEM MODEL

A. MODEL AND PROTOCOL INTRODUCTION

In this section, we study a multipoint FD SWIPT in WBAN. As shown in Fig. 1, the model consists of an AP and N sensors, S_i represents the i th sensor, $i = 1, 2, \dots, N$, and they all work in the same frequency band. The sensors are equipped with a TX antenna and a RX antenna, so that they can simultaneously transmit and receive signals over the same frequency band. The structure of FD sensors is shown in Fig. 2. We assume that the AP has a sufficient energy source, and the sensors themselves do not generate energy.

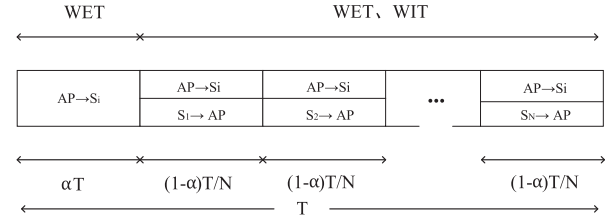


FIGURE 3. FD transmission protocol.

All of their energy comes from the AP during WET stage. We assume that the sensors do not store energy at the initial moment. We assume that h_i and g_i represent the channel parameters between AP and S_i and the distance between AP and S_i is represented by d_i . We assume that the channel state information (CSI) is ideal for both the AP and the sensors, that is, perfect CSI. The channels between each sensor and the AP are independent of each other, which conforms to the WBAN channel path loss model. The channel model is

$$PL = PL' + 10n\log_{10}\left(\frac{d_i}{d_0}\right), \quad (1)$$

where n is the path loss index, d_0 is the reference distance, and PL' is the path loss relative to the reference distance d_0 .

B. COMMUNICATION IN FD MODEL

Fig. 3 shows the energy and information transmission method based on the FD transmission protocol. The transmission protocol is divided into two parts, $(0, \alpha T)$ is the wireless energy transmission phase, and $(\alpha T, T)$ is the wireless information and energy simultaneous transmission phase which is divided into N time slots for N sensors. Therefore, the system includes a total of $(N + 1)$ time slots, T represents the block length. In the first αT wireless energy transmission phase, which is called the inactive phase, the AP broadcasts wireless energy to all sensor nodes. When the energy collected by each sensor reaches its threshold $E_{on,i}$, $i = 1, 2, \dots, N$, the next phase is started, and the remaining time $(1 - \alpha)T$ phase is called the active phase. Among them, the threshold is the minimum energy that can satisfy the sensors for information transmission. Different kinds of sensors have different energy thresholds. Therefore, it is required to start the active phase only when the energy collected by each sensor in the inactive phase is greater than its own energy threshold. In the active phase, the sensors transmit information through time division multiple access (TDMA), while the AP still broadcasts wireless energy, and the sensors receive the energy signal to realize energy accumulating. The main purpose of energy accumulation at this stage is to reduce the interruption probability of the system and increase the initial energy of the sensor at the next block. This reduces the time spent in the inactive phase of the next block and allocates more time to the active phase. For comparison, we consider a protocol with an AP in FD mode and sensors in HD mode, named as APFD. In this mode, the AP is equipped with two antennas, and the sensors are equipped with one antenna. In the $(i + 1)$ th time

slot, the sensor S_i can only send information signals to the AP, but can not receive energy signals from the AP.

In the communication of the FD model, we adopt the method of harvesting energy first and then sending information. Referring to the energy harvesting technology in wireless sensor networks, we consider that the transmit power of the AP is ρ_t and the distance between AP and S_i is R , then the power received by S_i is $\rho_i = \frac{\rho_t G_t G_r \lambda^2}{(4\pi)^2 R^2}$ [26], where G_t and G_r are the gains of the transmitting and receiving antennas, and λ is the wavelength. Here we only consider the received power of S_i , without considering the circuit details of its collected energy. In general theoretical research, the EH model is a linear function, but in practical applications, the nonlinear energy harvesting model is more in line with the requirements. Therefore, we consider the nonlinear energy harvesting model proposed in the paper [27]. The actual power used for energy harvesting at S_i is

$$\rho_i^{EH} = \frac{\psi_i - M\Omega_i}{1 - \Omega_i}, \quad (2)$$

$$\Omega_i = \frac{1}{1 + e^{ab}}, \quad (3)$$

$$\psi_i = \frac{M}{1 + e^{-a(\rho_i - b)}}, \quad (4)$$

where a and b are parameters related to the state of the EH circuit, ψ_i is a logical function related to ρ_i , M is the maximum power output by the EH circuit.

Therefore, the energy received by S_i in the inactive phase is

$$E_{0,i} = \rho_i^{EH} \alpha T, \quad (5)$$

where α is the time allocation ratio, $0 < \alpha < 1$. T is the time block.

In the active phase of simultaneous transmission of energy and information, due to the existence of interference such as self-interference, the efficiency of S_i receiving energy becomes low. In terms of reducing self-interference, it is mainly achieved through antenna cancellation technology. We place the TX antenna and the RX antenna at each end of the sensor. By using wireless channel attenuation between the receiving and transmitting antennas achieves self-interference cancellation. Because the AP transmitting power is large, the energy harvested by S_i from other sensors due to interference can be ignored. Let η_i be the energy conversion efficiency of S_i . Therefore, the energy received in the active phase S_i is

$$E_{1,i} = \rho_i^{EH} (1 - \alpha) \eta_i T. \quad (6)$$

Therefore, the total energy received by the entire time block of S_i is

$$E_i = E_{0,i} + E_{1,i} = \rho_i^{EH} \alpha T + \rho_i^{EH} (1 - \alpha) \eta_i T. \quad (7)$$

In the active phase, S_i will send information signals to the AP. The maximum transmit power that S_i can use is

$$P_i = \frac{E_i}{(1 - \alpha)T/N} = \frac{N\alpha}{1 - \alpha} \rho_i^{EH} + N\rho_i^{EH} \eta_i. \quad (8)$$

During the active phase, the total signal received by the AP from all sensors is

$$y = \sum_{i=1}^N \sqrt{P_{ii}} g_i x_i + n_0, \quad (9)$$

where P_{ii} represents the transmission power when S_i sends the information signals to the AP, x_i is the normalized information signal from the sensor, i.e., $E\{|x_i|^2\} = 1$, where $E\{\cdot\}$ stands for the expectation operator and $|\cdot|$ is the absolute value operator, and g_i represents the body surface channel gain from S_i to the AP, which is modeled by $h_i = g_i = \sqrt{10^{-\frac{-PL' - 10\log_{10}(\frac{d_i}{d_0})}{10}}}$ from formula (1), and $n_0 \sim CN(0, \sigma_0^2)$ represents the additive white Gaussian noise at the AP.

The signal-to-noise ratio of the received signal at the AP over the entire time block is

$$\gamma = \sum_{i=1}^N P_{ii} \frac{|g_i|^2}{\sigma_0^2}. \quad (10)$$

The standardized information throughput expression at the AP is

$$R(\alpha, P_{ii}) = \frac{(1 - \alpha)}{N} \log_2(1 + \gamma). \quad (11)$$

C. COMMUNICATION IN APFD MODEL

In the communication of the APFD model, the AP broadcasts energy to all sensors in the first inactive phase, and the sensors collect energy to reach their respective thresholds to start the active phase. In the active phase, because the sensor S_i works in the half duplex (HD) mode, in the $(i + 1)$ th slot, it only transmits the collected human physiological parameter information to the AP, and can not accept the energy information from the AP. In other time slots, S_i does not transmit information but can receive energy information from AP to achieve energy accumulation. Therefore, the total energy received by the entire time block of S_i in APFD is

$$E_i^{APFD} = \rho_i^{EH} \alpha T + \frac{(N - 1)}{N} \rho_i^{EH} (1 - \alpha) \eta_i T. \quad (12)$$

The maximum transmit power that S_i can use in the active phase is

$$P_i^{APFD} = \frac{E_i^{APFD}}{(1 - \alpha)T/N} = \frac{N\alpha}{1 - \alpha} \rho_i^{EH} + (N - 1)\rho_i^{EH} \eta_i. \quad (13)$$

In APFD model, the total signal that can be received by AP from all sensors in the active phase is

$$y^{APFD} = \sum_{i=1}^N \sqrt{P_{ii}^{APFD}} g_i x_i + n_0. \quad (14)$$

The signal-to-noise ratio of the received signal at the AP can be expressed as

$$\gamma^{APFD} = \sum_{i=1}^N P_{ii}^{APFD} \frac{|g_i|^2}{\sigma_0^2}. \quad (15)$$

The standardized information throughput at the AP can be expressed as

$$R^{APFD}(\alpha, P_{ii}^{APFD}) = \frac{(1-\alpha)}{N} \log_2(1 + \gamma^{APFD}). \quad (16)$$

III. OPTIMIZATION STRATEGY

In this part, in order to obtain the optimal average sum-throughput of the entire system, we jointly optimize the time allocation ratio α and the transmit power P_{ii} of S_i . The average sum-throughput optimization problem can be written as the following optimal problem

$$(P1): \max_{\alpha, P_{ii}} \frac{(1-\alpha)}{N} \log_2(1 + \sum_{i=1}^N P_{ii} \frac{|g_i|^2}{\sigma_0^2})$$

$$s.t. \quad C_1 \quad 0 < \alpha < 1,$$

$$C_2 \quad E_{0,i} \geq E_{on,i}, \quad i = 1, 2, \dots, N,$$

$$C_3 \quad P_{ii} = \min(P_i, P_{\max}). \quad (17)$$

In the above optimal problem, constraint C_1 denotes that the time allocation ratio should satisfy that $\alpha \in (0, 1)$, constraint C_2 indicates that in the inactive phase, the energy $E_{0,i}$ collected by S_i is required to be greater than its fixed energy threshold $E_{on,i}$. Substituting formula (5) into formula C_2 can obtain the requirements for S_i : $\alpha \geq \frac{E_{on,i}}{\rho_i T}$. In WBAN, the transmission power used by the sensors to transmit information can not exceed the maximum value that the human body can withstand, otherwise it will cause certain harm to the human body. Therefore, constraint C_3 means that the transmission power of the sensors when transmitting information is the smaller between the power collected during the whole phase and the maximum power the body can withstand.

To simplify the problem, for S_i we define $f(\alpha_i) = P_i - \frac{P_{\max}}{1-\alpha_i} \rho_i^{EH} + N \rho_i^{EH} \eta_i - P_{\max}$, $f(\alpha_i)$ to α_i first order inverse to get: $\frac{df(\alpha_i)}{d\alpha_i} = \frac{2\rho_i^{EH} N}{(1-\alpha_i)^2} > 0$, so we can prove that $f(\alpha_i)$ is a monotonically increasing function of α_i . Let $f(\alpha_i) = 0$, we can get: $\alpha_i = \frac{\rho_i^{EH} N}{N + \frac{P_{\max}}{\rho_i^{EH}} - \eta_i N}$. Therefore, it can be obtained that when $\alpha > \alpha_i$, $P_i > P_{\max}$, $P_{ii} = P_{\max}$, and when $\alpha < \alpha_i$, $P_i < P_{\max}$, $P_{ii} = P_i$. Therefore, when the optimal allocation time α is determined, the sensors can be divided into two parts, $\{S_1, S_2, \dots, S_M\}$ and $\{S_1, S_2, \dots, S_K\}$, where $M + K = N$. Among them, $S_m \in \{S_1, S_2, \dots, S_M\}$, $m = 1, 2, \dots, M$, satisfy $\alpha > \alpha_m$, that is $P_{im} = P_{\max}$, correspondingly, $S_k \in \{S_1, S_2, \dots, S_K\}$, $k = 1, 2, \dots, K$, satisfy $\alpha < \alpha_k$, that is $P_{ik} = P_{\max}$. According to the above derivation, the optimal problem (P1) can be transformed into the following form

$$(P2): \max_{\alpha} \frac{(1-\alpha)}{N} \log_2$$

$$\times (1 + \sum_{m=1}^M P_{\max} \frac{|g_m|^2}{\sigma_0^2} + \sum_{k=1}^K P_{\max} \frac{|g_k|^2}{\sigma_0^2})$$

$$s.t. \quad C_1 \quad 0 < \alpha < 1,$$

$$C_2 \quad \alpha \geq \frac{E_{on,i}}{\rho_i T}, \quad i = 1, 2, \dots, N. \quad (18)$$

By substituting formula (8) into (P2) and combining constraints C_1 and C_2 , we have

$$(P3): \max_{\alpha} \frac{(1-\alpha)}{N} \log_2(1 + \sum_{m=1}^M (\frac{N\alpha}{1-\alpha} \rho_m^{EH}$$

$$+ N \rho_m^{EH} \eta_m) \frac{|g_m|^2}{\sigma_0^2} + \sum_{k=1}^K P_{\max} \frac{|g_k|^2}{\sigma_0^2})$$

$$s.t. \quad C_1 \quad \frac{E_{on,i}}{\rho_i T} \leq \alpha < 1, \quad i = 1, 2, \dots, N. \quad (19)$$

We define that

$$A = 1 + \sum_{m=1}^M N \rho_m^{EH} \eta_m \frac{|g_m|^2}{\sigma_0^2} + \sum_{k=1}^K P_{\max} \frac{|g_k|^2}{\sigma_0^2},$$

$$B = \sum_{m=1}^M N \rho_m^{EH} \frac{|g_m|^2}{\sigma_0^2},$$

$$C = \max(\frac{E_{on,1}}{\rho_1 T}, \frac{E_{on,2}}{\rho_2 T}, \dots, \frac{E_{on,N}}{\rho_N T}). \quad (20)$$

Then the problem (P3) can be transformed into

$$(P4): \max_{\alpha} \frac{(1-\alpha)}{N} \log_2(A + B \frac{\alpha}{1-\alpha})$$

$$s.t. \quad C_1 \quad C \leq \alpha < 1. \quad (21)$$

From (P4) we can get that $R_{\max} = \frac{(1-\alpha)}{N} \log_2(A + B \frac{\alpha}{1-\alpha})$, the second-order inverse of $R_{\max}$ for α can be obtained: $\frac{d^2 R_{\max}}{d\alpha^2} = -\frac{B^2}{N(A - A\alpha + B\alpha)^2(1-\alpha) \ln 2}$. It can be proved that $R_{\max}$ is a convex function of α , so the problem (P4) is a convex optimization problem, which can be solved by applying convex optimization technology.

As for APFD model, the average sum-throughput optimization problem first can be written as problem (P1) and (P2). We don't repeat it here. Since the total energy that can be collected by the sensors in this mode is different from that in the full-duplex mode, so according to formula (13) we can get that the optimal problem can be shown as

$$(P5): \max_{\alpha} \frac{(1-\alpha)}{N} \log_2(1 + \sum_{m=1}^M (\frac{N\alpha}{1-\alpha} \rho_m^{EH}$$

$$+ (N-1) \rho_m^{EH} \eta_m) \frac{|g_m|^2}{\sigma_0^2} + \sum_{k=1}^K P_{\max} \frac{|g_k|^2}{\sigma_0^2})$$

$$s.t. \quad C_1 \quad \frac{E_{on,i}}{\rho_i T} \leq \alpha < 1, \quad i = 1, 2, \dots, N. \quad (22)$$

We define that

$$D = 1 + \sum_{m=1}^M (N-1) \rho_m^{EH} \eta_m \frac{|g_m|^2}{\sigma_0^2} + \sum_{k=1}^K P_{\max} \frac{|g_k|^2}{\sigma_0^2},$$

$$E = \sum_{m=1}^M N \rho_m^{EH} \frac{|g_m|^2}{\sigma_0^2},$$

$$F = \max(\frac{E_{on,1}}{\rho_1 T}, \frac{E_{on,2}}{\rho_2 T}, \dots, \frac{E_{on,N}}{\rho_N T}). \quad (23)$$

Then the problem (P5) can be transformed into

$$(P6) : \max_{\alpha} \frac{(1-\alpha)}{N} \log_2(D + E \frac{\alpha}{1-\alpha})$$

$$s.t. \quad C_1 \quad F \leq \alpha < 1. \quad (24)$$

We can get the second-order inverse of $R_{\max}^{APFD}$ for α is $\frac{d^2 R_{\max}^{APFD}}{d\alpha^2} = -\frac{E^2}{N(D-D\alpha+E\alpha)^2(1-\alpha)\ln 2}$. The result proves that $R_{\max}^{APFD}$ is a convex function of α too. So the problem (P6) is also a convex optimization problem, which can be solved by applying convex optimization technology.

From the above derivation, we can see that the difference between our proposed protocol and APFD is the maximum power that the sensor can emit. Since the sensors work in HD mode in APFD, the energy collected by the sensor in the information transmission phase is reduced by one time slot, so less energy can be collected. Ultimately lead to the difference between the optimal problems (P4) and (P6).

IV. SIMULATION RESULTS

In this section, we verify the previous formulations by comparing the average sum-throughput of the system in the FD mode and the average sum-throughput of the system in the APFD mode. Let's take a two-sensors system as an example. Due to the characteristics of WBAN, we must ensure the safety of the human body in the setting of parameters. Therefore, the setting of our simulation parameters is referred to [12]. In the body surface channel of the body area network, $PL' = 56.1dB$, $n = 5.8$, $d_0 = 10cm$ represents the reference distance, $d_1 = 25cm$ represents the distance between AP and S_1 , and $d_2 = 35cm$, represents the distance between AP and S_2 . We set the time block $T = 1s$, the energy threshold of S_1 is $E_{on,1} = 0.05mJ$, and the energy threshold of S_2 is $E_{on,2} = 0.04mJ$. The maximum transmission power $P_{\max} = 1.5mw$ that the human body can withstand, noise power at the AP is $\sigma_0^2 = -110dB$. To simplify the problem, we let $\eta_1 = \eta_2 = \dots = \eta_N = \eta$. The parameters of the nonlinear energy harvesting model are set according to [28] as: $a = 1500$, $b = 0.0026$, $M = 24mW$.

First of all, we compare the method proposed in this paper with the classical model of which both AP and sensors work in HD mode under different ρ_2 values, as shown in Fig. 4. We set ρ_1 is 0.5mw, η is 0.6 or 0.55 respectively. As we can see from the figure, the average sum-throughput of the system increase monotonically with an increase of ρ_2 . The reason is that with the increase of ρ_2 , the energy collected by S_2 keeps increasing, so that the power transmitted by S_2 in the active phase increases, which leads to an increase of the average sum-throughput of the system. Compared with the traditional method, this model can continue to collect energy in the active phase, which leads to the fact that S_2 can transmit more power when information is transmitted. So we can show that it works better than the traditional method. Next we will discuss the effect of the sensor's energy conversion efficiency, the sensor's energy harvesting rate, the impact of AP's noise power, the distance between sensors and AP and

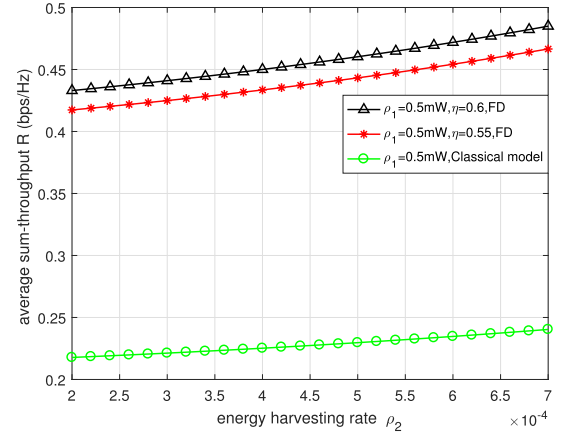


FIGURE 4. Comparison of average sum-throughput versus ρ_2 for FD mode and classical mode.

the computational complexity on average sum-throughput of the system.

Fig.5(a) shows the maximum average sum-throughput of the system under the optimal time allocation when the energy conversion efficiency η changes. At this time, ρ_1 is fixed at 0.5mw, and ρ_2 is 0.45mw or 0.4mw respectively. Fig. 5(b) shows the changing trend of the optimal time allocation of the system when η changes. It can be seen from the Fig. 5(a) that when η increases, the maximum average sum-throughput of the system increases monotonically, because $R_{\max}$ increases monotonically with the increase of η . While the optimal allocation time decreases monotonically with the increase of η . This is because when η is small, the transmission power of S_i is determined by the energy collected. It takes a large amount of time to allocate to the first stage to collect enough energy, and then the total average sum-throughput of the system reaches the maximum value. As η increases, in order to obtain a larger transmission power of S_i , the optimal allocation time will be reduced compared to the previous case. It can be seen from the figure that the larger ρ_2 is, the larger the system throughput is. This is because the greater the S_2 harvests energy rate is, the less time the system spends in the inactive state, and the more time can be allocated to the active state. It can also be proved from the figure that our protocol is better than APFD mode.

Fig. 6(a) shows the maximum average sum-throughput of the system under the optimal time allocation when the energy harvesting rate ρ_2 of S_2 changes. At this time ρ_1 is fixed at 0.5mw, and η is taken as 0.6 or 0.55 respectively. Fig. 6(b) shows the changing trend of the optimal time allocation of the system when ρ_2 changes. It can be seen from the figure that when ρ_2 increases, the maximum average sum-throughput of the system increase monotonically, and the optimal allocation time decreases as it increases. This is because when ρ_2 is small, the energy collected by S_2 during the entire stage is small. The power of the information that S_2 sends in the active stage is determined by the energy collected. In order to collect enough energy to reach the maximum throughput,

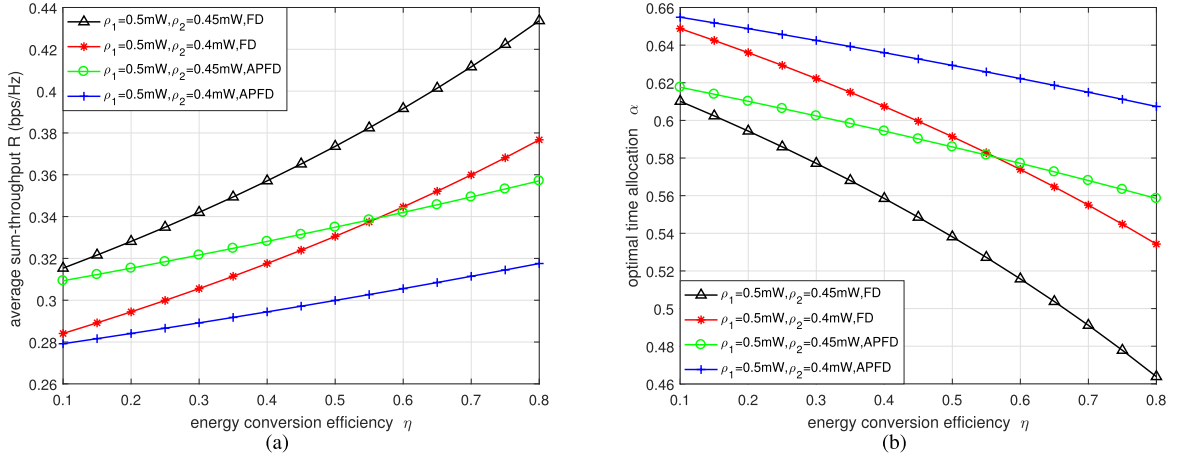


FIGURE 5. Average sum-throughput comparison of FD mode and APFD mode versus η . (a) The maximum average sum-throughput versus η , (b) The optimal time allocation versus η .

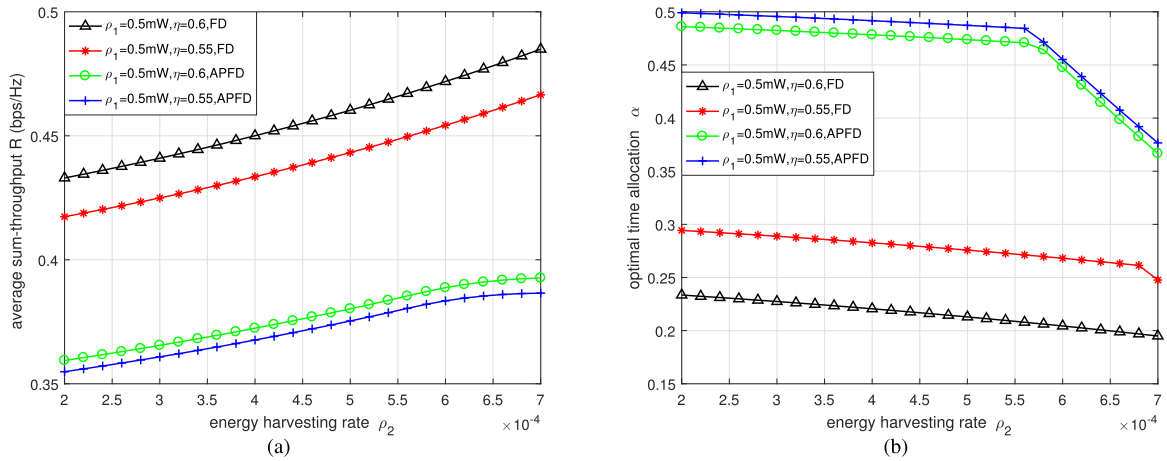


FIGURE 6. Average sum-throughput comparison of FD mode and APFD mode versus ρ_2 . (a) The maximum average sum-throughput versus ρ_2 , (b) The optimal time allocation versus ρ_2 .

more time should be allocated to the inactive phase. With the increase of ρ_2 , S_2 can collect more energy in a short time, and the power that can be transmitted during information transmission increases. When the value of ρ_2 increases to a certain degree, the power for S_2 to send information in the active phase is determined by $P_{\max}$, and it does not change anymore. The maximum average sum-throughput value tends to stabilize and reach the maximum value. It can also be seen from the figure that the throughput of the system increases with the increase of η . This is because the greater η is, the more energy S_i can collect, and the more time can be allocated to the active state.

Fig. 7 shows the average sum-throughput comparison of FD mode and APFD mode when d_i is variational. Fig. 7(a) shows the maximum average sum-throughput of the system when the distance d_1 is variational. Fig. 7(b) shows the maximum average sum-throughput of the system when the distance d_2 is variational. We set that $\rho_1 = 0.5\text{mW}$, $\eta = 0.6$, ρ_2 is 0.45mW or 0.4mW respectively. Obviously, the

maximum average sum-throughput decreases monotonically as d_i increases. This is because with the increase of d_i , the path loss between AP and S_i increases, resulting in the decrease of the strength of the energy signal that S_i receives and the information signal received by AP. By comparing Fig. 7(a) and Fig. 7(b), we can find that when the sensor is close to AP, changing the position of the sensor has a greater impact on the throughput of the system. The reason is that from formula (1) we can see that when the position of the sensor closer to the AP changes, it has a greater impact on the path loss. Although the performance difference between FD and APFD is small, the throughput of FD mode is larger than that of APFD mode.

As shown in Fig. 8, we also compare the system average sum-throughput of AP in FD mode and APFD mode when the additive noise power σ_0^2 at the AP changes. We set that $\rho_1 = 0.5\text{mW}$, $\eta = 0.6$, ρ_2 is 0.45mW or 0.4mW respectively. The figure shows that in the both mode the larger the noise power is, the smaller the average sum-throughput is, which is consistent with scientific facts. At the same time, it can be

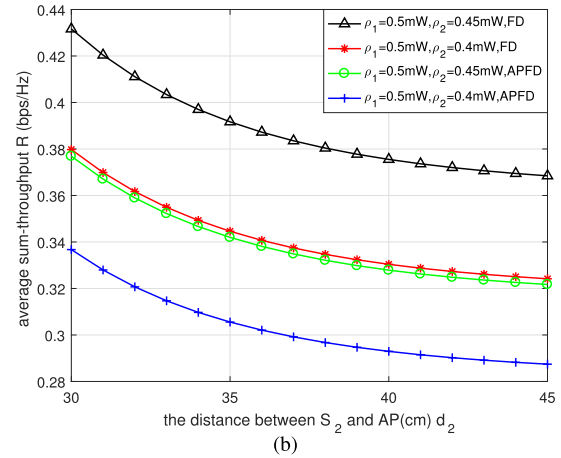
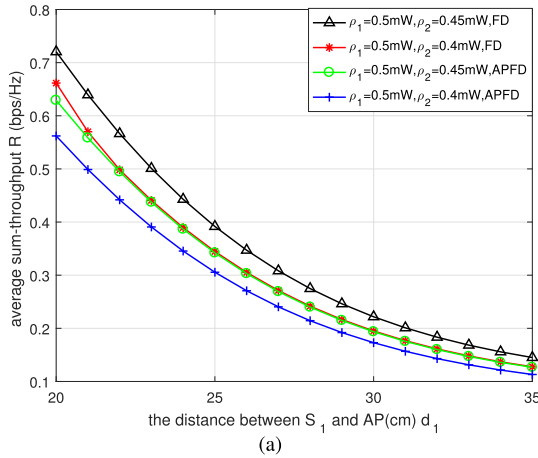


FIGURE 7. Average sum-throughput comparison of FD mode and APFD mode versus d_i . (a) The maximum average sum-throughput versus d_1 , (b) The maximum average sum-throughput versus d_2 .

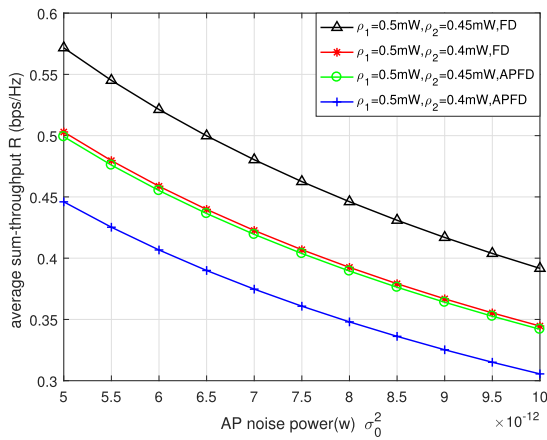


FIGURE 8. Comparison of average sum-throughput versus σ_0^2 for FD mode and APFD mode.

seen that when ρ_1 is fixed, the larger ρ_2 is, the greater the average sum-throughput is. We also verify that the system average sum-throughput in FD mode is greater than that in APFD mode, which proves that our proposed protocol performance is better.

In terms of computational complexity, (P4) and (P6) are both convex, we know that our proposed FD algorithm and APFD algorithm are guaranteed to converge, and the computational complexity of the two is the same. In addition, we record the running time of the two algorithms. The experiments are performed on MATLAB (R2016b) on a computer configured with an Intel Core i5-8250U CPU @ 1.60GHz and a NVIDIA GeForce MX150 GPU. Let's take the experiment in Fig. 5 as an example, run the FD protocol and APFD protocol on the computer and record the running time. We find that the average calculation time of the proposed protocol is 48.1082 seconds, while the average calculation time of the APFD protocol is 49.7146 seconds. This shows that the proposed protocol not only has better throughput

performance, but also has lower running time. Therefore, we obtain an algorithm with better throughput performance under the same computational complexity.

V. CONCLUSION

This paper studies a multi-point FD energy and information transmission problem based on TS allocation protocol in WBAN. In this protocol, the AP first broadcasts RF energy to the sensors with FD mode in the inactive phase, and then in the active phase, the sensors send back the collected physiological information parameters to the AP through TDMA, while simultaneously receiving energy information from the AP to realize energy accumulating. In the process of optimizing the time allocation ratio, we prove that the maximum average sum-throughput problem is convex. By using the convex optimization method, the optimal time allocation ratio can be obtained, thereby obtaining the maximum average sum-throughput. We also compare the case with the AP in FD and HD, and analyze the impact of different system parameters on system performance, such as sensor harvesting rate, energy conversion efficiency, noise power and the distance between sensor and AP. We also analyze the computational complexity of the two algorithms. Finally, the simulation results verify the rationality of the FD protocol proposed in this paper.

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